

Siyuan Mei

Ph.D. Student in Computer Science
Pattern Recognition Lab (LME), FAU Erlangen-Nürnberg
Martensstr. 3, 91058 Erlangen, Germany

siyuan.mei@fau.de
siyuan-mei.github.io
Google Scholar
github.com/siyuan-mei
LinkedIn

Research Interests

Medical AI: reconstruction, synthesis, segmentation, and registration.

Computer vision: diffusion models, generative modeling, representation learning, and dense prediction.

Vibe coding: software engineering workflows with large language models.

LLMs & agents: self-directed study of language modeling, reasoning, and agentic systems.

Education

FAU Erlangen-Nürnberg 2023–2027 (expected)

Ph.D. in Computer Science; Supervisor: Prof. Andreas Maier.

Peking University Health Science Center May–Oct. 2026

Visiting Scholar, Artificial Intelligence for Medical Imaging (AIMI) Lab; Supervisor: Prof. Yixing Huang.

FAU Erlangen-Nürnberg 2020–2023

M.Sc. in Medical Engineering; thesis grade: 1.0.

Zhejiang University of Science and Technology 2016–2020

B.Eng. in Automation; honored graduate scholarship.

Professional Experience

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) 2023–2027

Ph.D. Student, Pattern Recognition Lab; Supervisor: Prof. Andreas Maier

- Fully funded through the ERC project *Advancing osteoporosis medicine by observing bone microstructure and remodelling using a 4D nanoscope*; developed **BigReg**, a registration pipeline for high-resolution X-ray microscopy and light-sheet fluorescence microscopy.
- Research on medical AI and computer vision, including adaptation of large-scale visual models and generative medical imaging with GANs, diffusion models, and flow-matching models.
- Teaching: *Flat-panel CT Reconstruction* and *Vibe Coding: Software Engineering with LLMs*.
- Outcomes: publications in Pattern Recognition, Journal of Medical Imaging, IEEE TMI, MICCAI, and related venues.

Peking University Health Science Center, AIMI Lab May–Oct. 2026

Visiting Scholar; Supervisor: Prof. Yixing Huang

Artificial Intelligence for Medical Imaging Lab, Institute of Medical Technology, Peking University Health Science Center, 240 Pathology Building, No. 38 Xueyuan Road, Haidian District, Beijing, China.

- Participated in the MICCAI BIC-MAC Challenge and trained a multimodal-to-CT generative model for PET attenuation correction; ranked first on the validation-set CT metric.
- Received training in clinical CT applications, imaging workflows, and related technical practice.
- Visited the ICT Center in Chongqing and attended the MISC 2026 conference at Tsinghua University.

Honors & Awards

- **1st place, Task 2** and **4th place, Task 1**, MICCAI SynthRAD2025 Challenge, 2025.
- Fully funded Ph.D. through the ERC 4D Nanoscope project, FAU, 2023–2027.

Selected Publications

1. Siyuan Mei et al. **Time Matters: Rethinking Diffusion and Flow Models in One-Step Medical Image Translation.** *MICCAI 2026*, Early Accept; Oral Presentation; first author.
2. Mengchen Fan et al., including Siyuan Mei. **Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation.** *MICCAI 2026*, Early Accept; Spotlight Presentation.
3. Daiqi Liu et al., including Siyuan Mei. **Speech-Guided Multimodal Learning for Vocal Tract Segmentation in Real-Time MRI.** *MICCAI 2026*, Early Accept; Oral Presentation; [arXiv](#).
4. Siyuan Mei et al. **WING: A Window-Prior-Based Generative Network with Gated Inception for Cross-Modality CT Synthesis.** *MICCAI MIART Workshop 2026*, accepted workshop paper; [code](#).

5. Siyuan Mei et al. **Vision Transformer Hook for Dense Predictions**. *Pattern Recognition* 179 (2026), 113818, DOI; first author.
6. Siyuan Mei et al. **BigReg: An Efficient Registration Pipeline for High-Resolution X-Ray and Light-Sheet Fluorescence Microscopy**. *Journal of Medical Imaging* 12(5), 054004 (2025), DOI; Featured Paper; first author.

Other Publications

1. Siyuan Mei, co-author. **TCCT: Trajectory-Conditioned CBCT Reconstruction for Sinusoidal Non-Circular Orbits with a Fourier Neural Operator**. *MICCAI 2026 BIC-MAC Challenge*, Poster.
2. Daiqi Liu et al., including Siyuan Mei. **Speech-Guided Multimodal Learning for Vocal Tract Segmentation in Real-Time MRI**. *MICCAI 2026*, Oral Presentation; [arXiv](#).
3. Y. Sun et al., including Siyuan Mei. **Filter2Noise: A Framework for Interpretable and Zero-Shot Low-Dose CT Image Denoising**. *Journal of Medical Imaging* 13, 024004 (2026), DOI.
4. C. Ye et al., including Siyuan Mei. **Robustness and Stability Analysis of Differentiable Shift-Variant FBP for Cone-Beam CT under Challenging Acquisition Settings**. *Machine Learning for Biomedical Imaging* (2026), DOI.
5. M. Thies et al., including Siyuan Mei. **A Gradient-Based Approach to Fast and Accurate Head Motion Compensation in Cone-Beam CT**. *IEEE Transactions on Medical Imaging* (2024), DOI.
6. C. Ye et al., including Siyuan Mei. **DRACO: Differentiable Reconstruction for Arbitrary CBCT Orbits**. *Physics in Medicine and Biology* 70, 075005 (2025), DOI.
7. M. Gu et al., including Siyuan Mei. **Unsupervised Domain Adaptation Using Soft-Labeled Contrastive Learning with Reversed Monte Carlo Method for Cardiac Image Segmentation**. *MICCAI 2024*, DOI.
8. L. Pfaff et al., including Siyuan Mei. **No-New-Denoiser: A Critical Analysis of Diffusion Models for Medical Image Denoising**. *MICCAI 2024*, [open access](#).
9. Zirong Li, Siyuan Mei et al.. **Generative Drifting for Conditional Medical Image Generation**. *arXiv preprint* (2026), [arXiv](#).
10. Siyuan Mei, F. Fan, A. Maier. **Metal Inpainting in CBCT Projections Using Score-based Generative Model**. *IEEE ISBI 2023*, DOI.
11. Siyuan Mei, Y. Xia, F. Fan. **GANeXt: A Fully ConvNeXt-Enhanced Generative Adversarial Network for MRI- and CBCT-to-CT Synthesis**. *arXiv preprint* (2025), [arXiv](#).

Teaching

Flat-panel CT Reconstruction Summer semesters 2024–2026.	FAU
Vibe Coding: Software Engineering with LLMs Summer semester 2026.	FAU

Professional Service

Technical Editor: Journal of Medical Imaging.

Journal Reviewer: Pattern Recognition; Journal of Machine Learning for Biomedical Imaging (MELBA).

Conference Reviewer: MICCAI.

Open-source and Teaching Resources

WING, JiR, and ViT-Hook; [Vibe Coding Slides](#) (Springer book companion).